,

€ • , • *f*

,

„ , …

6

-

† ‡ :

„^ …

‰ ‡ :

. Š ‰2402 _____ ‹ ,Œ 06

† :

‡ ‡ _____ ‰ ‰. .

2026

€

• , ‡

, ‡ . ,Œ

‡ , Ž•• ‡

‡

„ ‘ ‡ -2….

’ ( ) N = 12, ,Œ

• “ 06. † ,Œ • 6 “

,Œ , ‡ ” ‡ 2 ,

•,ƒ„ …† ‡ ,‡ ^ ‰ (Š).

• ‡ Ž•• ‡

‡ ‡ α_1 = 0,05, ‡

‡ • α_2 = 0,01.

$$F = 10000 + 50 - N = 10000 + 50 - 12 = 10600 \text{ ~}$$

$$F_1 = 2100 + 35 - N = 2100 + 35 - 12 = 2520 \text{ ~}$$

$$F_3 = 705 + 25 - N = 705 + 25 - 12 = 1005 \text{ ~}$$

1 ™ • 12 / ,Œ 06

†ƒ,^,	‹ ^† ,†	^† ,†
’	N	12
’ ,Œ •	“	06
‡	“	‰•
•	F	10600 ~
~ ‡ ‡ ‡	F1	2520 ~
~ ‡ •	F3	1005 ~
• ‡ ††	–1	0,05
• ‡ †	–2	0,01

,‡ , 0 š f š F_3 |H(f)| š

0,01, ‡ F_1 š f š F /2 •

‡ 0,95 š |H(f)| š 1,05. ‰

, “ , ’ F /2 = 5300 ~ .

Œ Š •

1.1 Š ‰ ‰ †‹ , ,Ž† ƒ„ • Š

• Ž•• ‡
‡ ‡ • .
’ , , f_c ,
, ‡ ” 2F .

$$F_c=(F_1+F_3)/2=(2520+1005)/2=1762,5\ \tilde{}$$

$$f_c=(F_1+F_3)/2F_{\ }=(2520+1005)/2\text{-}10600=0,166273584906$$

• ‰• ‡ , ‡
’• ‡ ‡ € › . † ” ,Œ
, •• :

$$h_u(0)=1\text{œ}2f_c=0,667452830189$$

$$h_u(n)=\text{œ}\sin(2\ f_c n)/\ n, n\bullet 0$$

• , •• - , €
•• • , , , ,
,Œ • € €.

1.2 • ‘ ’† … “ ”‹† ,‘ Ž“ ,Žƒ, • •,ƒ„

‡ ‡
• ƒ ‡ ‡ :

$$a_{min}=\text{œ}20\lg(\ _2)=\text{œ}20\lg(0,01)=40\ \text{Š}$$

$$_F=(F_1\text{œ}F_3)/F_{\ }=(2520\text{œ}1005)/10600=0,142924528302$$

a_{min} = 40 Š ž 21 Š, ‡ ‡
D ‡ R , € ‡ (5)™(7)
, :

$$D=(a_{min}\text{œ}7,95)/14,36=(40\text{œ}7,95)/14,36=2,231894150418$$

$$R=\text{Ÿ}D/\ _F+1\ =\text{Ÿ}16,615893065630\ =16$$

$$N=R+1=17$$

• ‡ ¡ ‡ 21 š a_{min} š 50 Š
:

$$=0,5842(a_{min}\text{œ}21)^{0,4}+0,07886(a_{min}\text{œ}21)=3,395321052261$$

† , ‡ R = 16 ,Œ , N = 17 ,Œ . ¢ ‡ ‡ , •• - I ‡ , ‡ ‰• . ,Œ , € œ8 š n š 8:

$$w(n) = I_0(\pounds(1\text{œ}(2n/(N\text{œ}1))^2))/I_0(\ )$$

$I_0(x)$ “ Š ‡ ‡ . ‰ w(0) = 1, , , : w(œn) = w(n). •• ‡ ‡ $h(n) = h_u(n)$-w(n).

1.3 •••,–,† • † *f„* • •,*f„*

,Œ ‡ 2. ‡ , •• ,Œ : b(i) = h(n), n = i œ 8, i = 0,¤,16. ‰ , ; ,Œ ‡ ‡ ‡ , €.

2 ™ ,Œ •• , ” ‰•

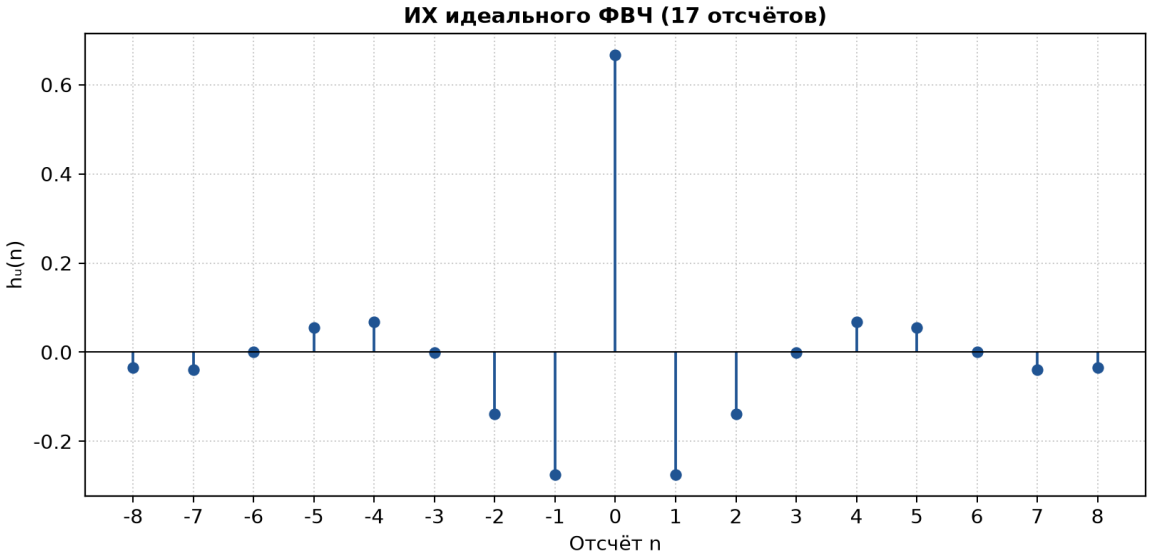
•••,–,†	n	h–(n)	w(n)	h(n) = b(i)
b(0)	-8	œ0,03484439	0,14796535	œ0,00515576
b(1)	-7	œ0,03898169	0,26611301	œ0,01037353
b(2)	-6	0,00078613	0,40226828	0,00031624
b(3)	-5	0,05552176	0,54681505	0,03036013
b(4)	-4	0,06851967	0,68826332	0,04715958
b(5)	-3	œ0,00078616	0,81449893	œ0,00064032
b(6)	-2	œ0,13822362	0,91416844	œ0,12635967
b(7)	-1	œ0,27527053	0,97801702	œ0,26921926
b(8)	0	0,66745283	1,00000000	0,66745283
b(9)	1	œ0,27527053	0,97801702	œ0,26921926
b(10)	2	œ0,13822362	0,91416844	œ0,12635967
b(11)	3	œ0,00078616	0,81449893	œ0,00064032
b(12)	4	0,06851967	0,68826332	0,04715958
b(13)	5	0,05552176	0,54681505	0,03036013
b(14)	6	0,00078613	0,40226828	0,00031624

• • • , – , †	n	h–(n)	w(n)	h(n) = b(i)
b(15)	7	œ0,03898169	0,26611301	œ0,01037353
b(16)	8	œ0,03484439	0,14796535	œ0,00515576

” , b(8) = 0,66745283, ‡ ” € €

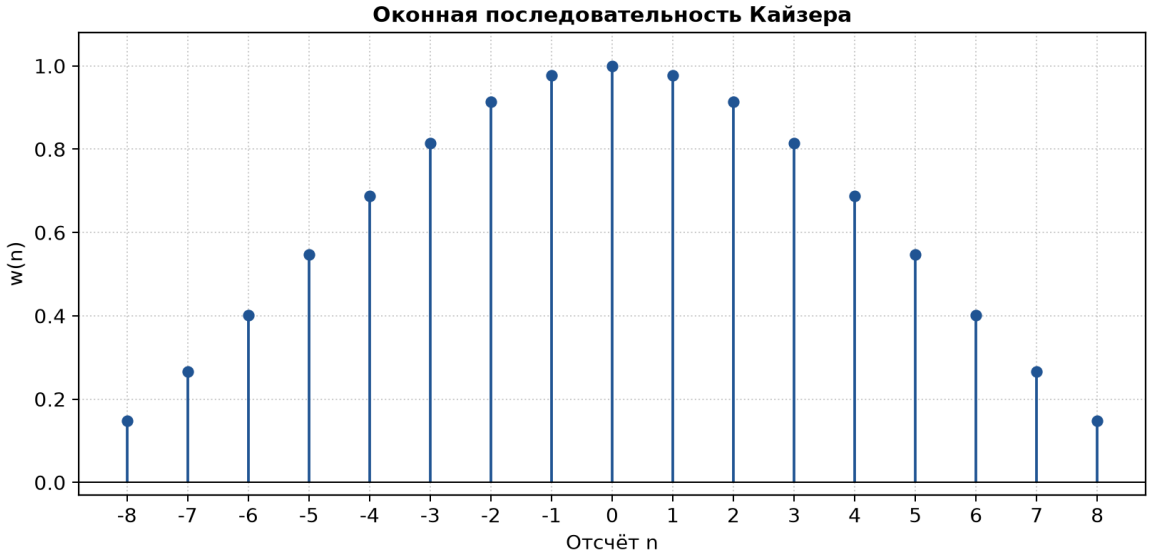
, ‡‡ • ‡

R/2 = 8 ,Œ .



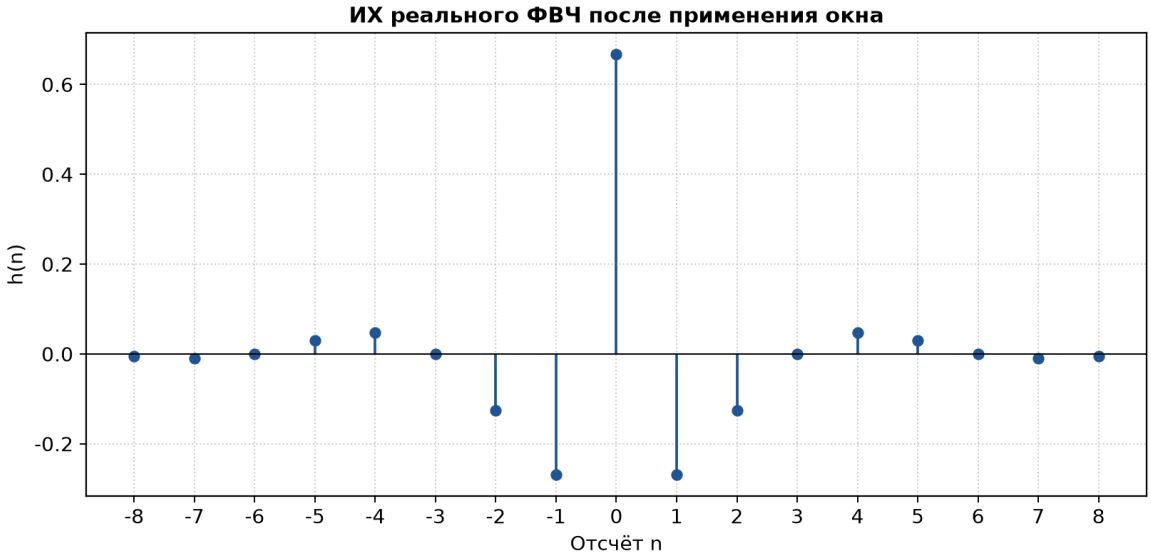
1 ™ • • ‰ • , , €

,Œ



2 ™ ‡ ‡ ¡ =

3,39532105



3 ™ ˆ • • ‰ • ‡ •

1.4 † †Ž ˆ • ~ “ – , , ‹ ‰ † ~ … † , †

† • • ,Œ ‡ ,

• •- 17. ¥ ‡ ,

€ :

$$H(z) = \sum_{i=0}^{16} b_i z^{-i}$$

$$y(k) = \sum_{i=0}^{16} b_i x(k-i)$$

‰ 16 ‡ Œ

• , 17 ”

b(0), ¤ ,b(16) 17 .

” € , 2;

› ” • .

Š •

2.1 *f* „ • † ˆ “ , Š

Ž • • , ‡ ‡ ‡ ,

” *f* 0,01 ~ .

‡ a(f) = -20 lg |H(f)|.

3 ™ ,Œ , Ž • •

f, ™ –	|H(f)|	a(f), Ž €	’ • ‰ *f*
0,0	0,000372377	68,580348	† € ›
1005,0	0,009669005	40,292364	~ † F3

f, ™–	|H(f)|	a(f), Ž€	’ •‰ƒ
1762,5	0,500312735	6,015169	• Fc
2520,0	0,990391456	0,083862	~ †† F1
5300,0	0,999119558	0,007651	• ’

’ , |H(1762,5)| = 0,500312735 § 0,5, ,
, Ž••. ’
† 40,292364 Š,
40 Š ‡ , ‡ 0,292364 Š.

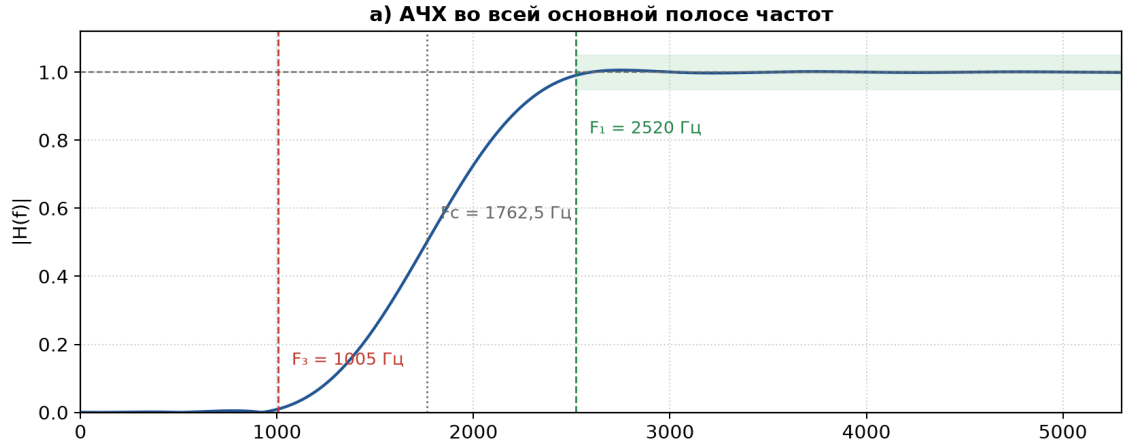
2.2 •‘ ƒ † ,† Ž ‘~‰“ … , ’ , , ’ ƒ„ ‰ „ ‘ Ž“

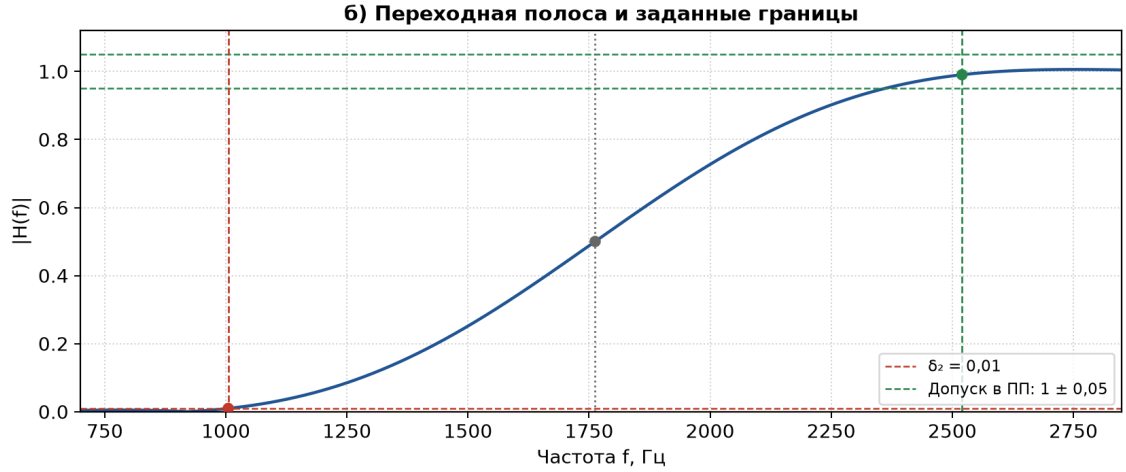
Ž•• ‡ • 0¤1005 ~
0,009669005 1005 ~ :
0,009669005 š $–_2$ = 0,01. ‰ ‡ ‡ ‡ 2520¤5300
~ ‡ , ‡ 0,990391456¤1,005886154,
• ‡ 0,95¤1,05.
††
0,009608544 š $–_1$ = 0,05.

4 ™ ‘ ,Œ ‡

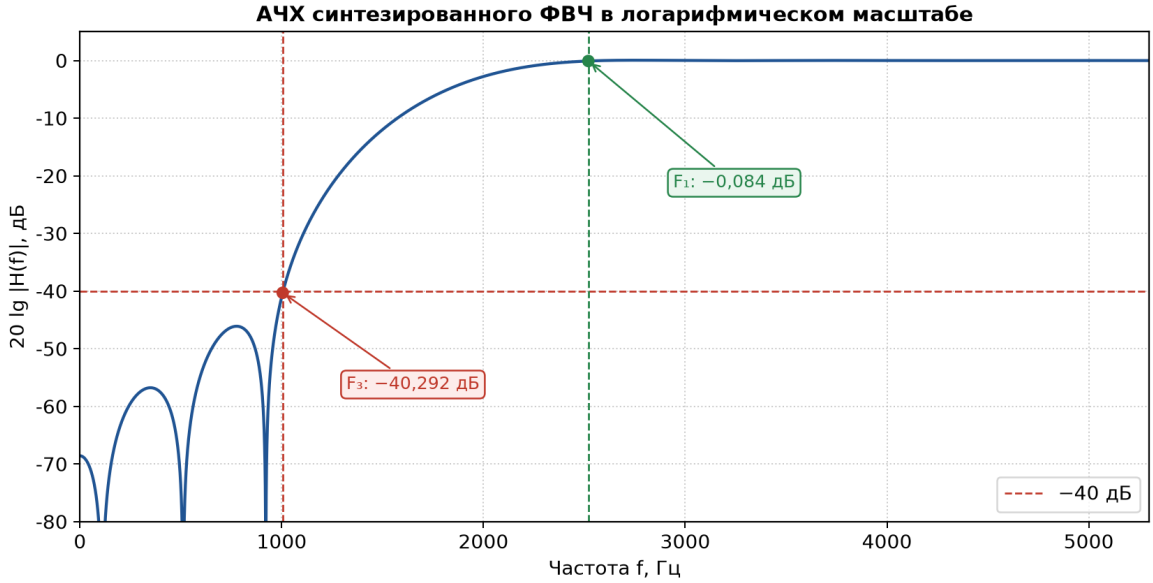
R	N	max |H| …	min |H| …	max |H| …	† … ,
12	13	0,05879905	0,93728946	1,00836248	
14	15	0,03051144	0,97288867	1,00701934	
16	17	0,00966900	0,99039146	1,00588615	

† R = 14 † ‡ 0,030511, ,
ƒ 0,01. ‰ ƒ ,Œ ‡ • ‡
‡ , ‡ R = 16 ‡ € .
‘ , R = 16 “ ‡ › ,Œ
‡ ¡.





4 ™ Ž • • ‰ • *f* ‡ ‡



5 ™ Ž • • ‰ • , *f*

• ‡ ‡ , scipy.signal.firwin

• N = 17, F_c = 1762,5 ~ i ,

” : €

1.086e-16. † b(i)

† 0,009668994, ‡ † † “

0,990391443 ¤ 1,005886157; € .

š › Š

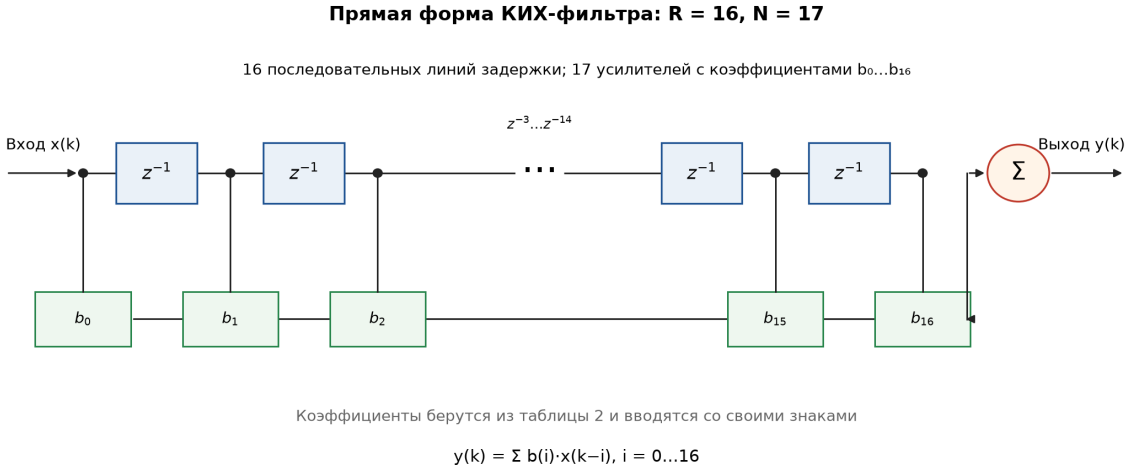
• ‡ ‡

€ • •- . ,Œ

4™7 € ” ‡

„ ‘ ‡ -2… ‡ € *f*

‡ .



6 ™ †

• •-



7 ™ , ‡

† 17 ,Œ , ‡
‡ € b(0),¤,b(16), ‡ , •• €. ¢
‡ • , € ••. ‡
€ ,Œ ,
‡ ‡‡
• 8/F § 0,754717 .

3.1 Ž “ ‘ …† “ , … ‰ †Ž† ‘†“ -2

1. ‰ „ ··· ¨ „ ‘ ···, ,
F = 10600 ˜ . ‰ , ‡ 16
• ‡ .
2. † € , 17
• 6. ‰ ”
b(0),¤,b(16) 2 ‡ • TXT-
. ‰ ‡ 17
.
3. • ˜ •: ‡ “ ‡ , ,
f ‡ “ 1. † € ,
‡ , ‡
•• Ž••
, *f* .
4. ‘
, : 1005 ˜ |H| § 0,009669 (a § 40,292 Š);
1762,5 ˜ |H| § 0,500313; 2520 ˜ |H| § 0,990391; 5300
˜ |H| § 0,999120. † Ž•• , , 1005 ˜
‡ *f* 0,01, ‡ 2520 ˜
‡ *f* 0,05.

• 12 ,Œ • 06
‰• . †
F = 10600 ˜ , F_1 = 2520 ˜ F_3 = 1005 ˜
, F_c = 1762,5 ˜ . † : a_{min} = 40 Š,
D = 2,23189415, ¡ = 3,39532105. † ,
,Œ ‡ R = 16, N = 17 17 ,
” .
,Œ Ž•• €: max |H| † =
0,009669005 š 0,01, ‡ Ž•• †† =
0,990391456 ¤ 1,005886154 ‡ ‡ 0,95 ¤ 1,05.
€ •• ‡‡ €
• 8 ,Œ . • „ ‘ ‡ -2…
‡ ‡ , ”
, .

Š

1. © ¥. ., ‘ ¥. ., • € .˜. ˆ
. © ‹ 6: ‘
••- ‡
. .: ª ‘ •, 2016.
2. ‘ Ž.Š. ˆ : ,
‡ . 3- . ‘ † .: Š•‰-† , 2011. 768 .
3. ‡‡ Ž., « . ˆ
. 2- . .: , 2007.